

#In this code we will demonstrate proposition 7.22. We will verify that there is a single configuration along the family described by γ_4 with zero total mass.

#Define the mass component ni , nj and nk

$$ni := \frac{y}{8} \cdot (s^{-3} - T) \cdot (z^{-3} - w^{-3}) :$$

$$nj := \frac{x}{8} \cdot (s^{-3} - T) \cdot (z^{-3} + w^{-3} - 2 \cdot x^{-3}) :$$

$$nk := \frac{x}{4} \cdot ((w^{-3} - x^{-3}) \cdot (y^{-3} - z^{-3}) + (z^{-3} - x^{-3}) \cdot (y^{-3} - w^{-3})) :$$

#Define the total mass $m0$

$$m0 := 2 \, ni + 2 \, nj + nk$$

$$m0 := \frac{y \left(\frac{1}{s^3} - T \right) \left(\frac{1}{z^3} - \frac{1}{w^3} \right)}{4} + \frac{x \left(\frac{1}{s^3} - T \right) \left(\frac{1}{z^3} + \frac{1}{w^3} - \frac{2}{x^3} \right)}{4} \quad (1)$$

$$+ \frac{x \left(\left(\frac{1}{w^3} - \frac{1}{x^3} \right) \left(\frac{1}{y^3} - \frac{1}{z^3} \right) + \left(\frac{1}{z^3} - \frac{1}{x^3} \right) \left(\frac{1}{y^3} - \frac{1}{w^3} \right) \right)}{4}$$

#The configurations described by γ_4 satisfy`

$$s := \frac{y}{2}; T := \frac{z^{-3} + w^{-3}}{2}$$

$$s := \frac{y}{2}$$

$$T := \frac{1}{2 z^3} + \frac{1}{2 w^3} \quad (2)$$

#Rewriting $m0$

$$m0$$

$$\frac{y \left(\frac{8}{y^3} - \frac{1}{2 z^3} - \frac{1}{2 w^3} \right) \left(\frac{1}{z^3} - \frac{1}{w^3} \right)}{4} + \frac{x \left(\frac{8}{y^3} - \frac{1}{2 z^3} - \frac{1}{2 w^3} \right) \left(\frac{1}{z^3} + \frac{1}{w^3} - \frac{2}{x^3} \right)}{4} \quad (3)$$

$$+ \frac{x \left(\left(\frac{1}{w^3} - \frac{1}{x^3} \right) \left(\frac{1}{y^3} - \frac{1}{z^3} \right) + \left(\frac{1}{z^3} - \frac{1}{x^3} \right) \left(\frac{1}{y^3} - \frac{1}{w^3} \right) \right)}{4}$$

#Collecting the numerator and denominator of $m0$

$$\text{numer}(m0); \text{denom}(m0)$$

$$-x \left(w^6 x^3 y^3 - 18 w^6 x^3 z^3 + w^6 x^2 y^4 - 16 w^6 x^2 y z^3 - 4 w^6 y^3 z^3 + 36 w^6 z^6 + 6 w^3 x^3 y^3 z^3 \right)$$

$$-18 w^3 x^3 z^6 + 16 w^3 x^2 y z^6 - 4 w^3 y^3 z^6 + x^3 y^3 z^6 - x^2 y^4 z^6) \\ 8 y^3 z^6 w^6 x^3$$

(4)

#Applying rational parameterization

$$\text{parametrization} := \left\{ w = a + \frac{1}{a}, \right.$$

$$z = b + \frac{1}{b},$$

$$x = a - \frac{1}{a} + b - \frac{1}{b},$$

$$y = a - \frac{1}{a} - b + \frac{1}{b} \Big\} :$$

simplify((expand(subs(parametrization, numer(m0)))));

simplify((expand(subs(parametrization, denom(m0)))));

$$-\frac{1}{a^{13} b^{13}} (2 (a b - 1) ((-28 b^{16} - 66 b^{14} + 8 b^{12} + 50 b^{10} + 30 b^8) a^4 + (20 b^{15} + 73 b^{13} \\ + 59 b^{11} + 24 b^9) a^3 + (-2 b^{14} + 4 b^{12} - 2 b^{10}) a^2 + (b^{13} - b^{11}) a + (-b^{13} + b^{11}) a^{23} \\ + (-2 b^{14} + 4 b^{12} - 2 b^{10}) a^{22} + (-14 b^{15} - 55 b^{13} - 41 b^{11} - 18 b^9) a^{21} + (-28 b^{16} \\ - 66 b^{14} + 8 b^{12} + 50 b^{10} + 30 b^8) a^{20} + (-35 b^{17} - 89 b^{15} - 153 b^{13} - 107 b^{11} - 111 b^9 \\ - 33 b^7) a^{19} + (16 b^{18} - 22 b^{16} - 20 b^{14} + 312 b^{12} + 450 b^{10} + 210 b^8 + 14 b^6) a^{18} + (-3 b^{19} - 190 b^{17} - 226 b^{15} + 154 b^{13} + 364 b^{11} - 60 b^9 - 60 b^7 + 21 b^5) a^{17} + (-42 b^{20} \\ - 54 b^{18} + 368 b^{16} + 1538 b^{14} + 2587 b^{12} + 2038 b^{10} + 600 b^8 - 66 b^6 - 42 b^4) a^{16} + (-20 b^{21} - 51 b^{19} - 306 b^{17} - 70 b^{15} + 929 b^{13} + 1009 b^{11} - 76 b^9 - 108 b^7 + 87 b^5 + 14 b^3) \\ a^{15} + (-b^{24} - 4 b^{22} - 8 b^{20} + 368 b^{18} + 2433 b^{16} + 5952 b^{14} + 7200 b^{12} + 4912 b^{10} \\ + 1853 b^8 + 328 b^6 - 8 b^4 - 4 b^2 - 1) a^{12} + (-16 b^{21} - 29 b^{19} + 212 b^{17} + 76 b^{15} \\ - 753 b^{13} - 933 b^{11} + 70 b^9 + 274 b^7 + 65 b^5 + 10 b^3) a^9 + (40 b^{20} + 274 b^{18} + 696 b^{16} \\ + 1210 b^{14} + 1767 b^{12} + 1710 b^{10} + 928 b^8 + 262 b^6 + 40 b^4) a^8 + (-5 b^{19} + 60 b^{17} \\ - 44 b^{15} - 266 b^{13} - 216 b^{11} + 258 b^9 + 190 b^7 + 23 b^5) a^7 + (22 b^{18} + 14 b^{16} + 70 b^{14} \\ + 432 b^{12} + 540 b^{10} + 246 b^8 + 20 b^6) a^6 + (17 b^{17} + 53 b^{15} + 135 b^{13} + 89 b^{11} + 75 b^9 \\ + 15 b^7) a^5 - b (b^{22} + 61 b^{20} + 123 b^{18} - 206 b^{16} - 477 b^{14} + 192 b^{12} + 992 b^{10} + 581 b^8 \\ - 26 b^6 - 141 b^4 - 43 b^2 - 1) a^{11} + 2 b^2 (b^{20} + 45 b^{18} + 307 b^{16} + 1003 b^{14} + 1887 b^{12} \\ + 2184 b^{10} + 1700 b^8 + 887 b^6 + 292 b^4 + 45 b^2 + 1) a^{10} + 2 b^2 (b^{20} - 37 b^{18} - 12 b^{16} \\ + 729 b^{14} + 2350 b^{12} + 3184 b^{10} + 2163 b^8 + 613 b^6 - 27 b^4 - 37 b^2 + 1) a^{14} + b (b^{22} \\ - 47 b^{20} - 77 b^{18} + 36 b^{16} + 585 b^{14} + 992 b^{12} + 192 b^{10} - 733 b^8 - 304 b^6 + 95 b^4 + 29 b^2 \\ - 1) a^{13} + a^{24} b^{12} + b^{12}) (b + a))$$

$$\frac{8 (b^2 + 1)^6 (a^2 + 1)^6 (ab - 1)^3 (ab + 1)^3 (-b + a)^3 (b + a)^3}{a^{12} b^{12}} \quad (5)$$

#Define m0

$$\begin{aligned} m0 := & ((-28 b^{16} - 66 b^{14} + 8 b^{12} + 50 b^{10} + 30 b^8) a^4 + (20 b^{15} + 73 b^{13} + 59 b^{11} + 24 b^9) a^3 \\ & + (-2 b^{14} + 4 b^{12} - 2 b^{10}) a^2 + (b^{13} - b^{11}) a + (-b^{13} + b^{11}) a^{23} + (-2 b^{14} + 4 b^{12} \\ & - 2 b^{10}) a^{22} + (-14 b^{15} - 55 b^{13} - 41 b^{11} - 18 b^9) a^{21} + (-28 b^{16} - 66 b^{14} + 8 b^{12} + 50 b^{10} \\ & + 30 b^8) a^{20} + (-35 b^{17} - 89 b^{15} - 153 b^{13} - 107 b^{11} - 111 b^9 - 33 b^7) a^{19} + (16 b^{18} \\ & - 22 b^{16} - 20 b^{14} + 312 b^{12} + 450 b^{10} + 210 b^8 + 14 b^6) a^{18} + (-3 b^{19} - 190 b^{17} - 226 b^{15} \\ & + 154 b^{13} + 364 b^{11} - 60 b^9 - 60 b^7 + 21 b^5) a^{17} + (-42 b^{20} - 54 b^{18} + 368 b^{16} + 1538 b^{14} \\ & + 2587 b^{12} + 2038 b^{10} + 600 b^8 - 66 b^6 - 42 b^4) a^{16} + (-20 b^{21} - 51 b^{19} - 306 b^{17} - 70 b^{15} \\ & + 929 b^{13} + 1009 b^{11} - 76 b^9 - 108 b^7 + 87 b^5 + 14 b^3) a^{15} + (-b^{24} - 4 b^{22} - 8 b^{20} + 368 b^{18} \\ & + 2433 b^{16} + 5952 b^{14} + 7200 b^{12} + 4912 b^{10} + 1853 b^8 + 328 b^6 - 8 b^4 - 4 b^2 - 1) a^{12} + (\\ & -16 b^{21} - 29 b^{19} + 212 b^{17} + 76 b^{15} - 753 b^{13} - 933 b^{11} + 70 b^9 + 274 b^7 + 65 b^5 + 10 b^3) a^9 \\ & + (40 b^{20} + 274 b^{18} + 696 b^{16} + 1210 b^{14} + 1767 b^{12} + 1710 b^{10} + 928 b^8 + 262 b^6 + 40 b^4) a^8 \\ & + (-5 b^{19} + 60 b^{17} - 44 b^{15} - 266 b^{13} - 216 b^{11} + 258 b^9 + 190 b^7 + 23 b^5) a^7 + (22 b^{18} \\ & + 14 b^{16} + 70 b^{14} + 432 b^{12} + 540 b^{10} + 246 b^8 + 20 b^6) a^6 + (17 b^{17} + 53 b^{15} + 135 b^{13} \\ & + 89 b^{11} + 75 b^9 + 15 b^7) a^5 - b (b^{22} + 61 b^{20} + 123 b^{18} - 206 b^{16} - 477 b^{14} + 192 b^{12} \\ & + 992 b^{10} + 581 b^8 - 26 b^6 - 141 b^4 - 43 b^2 - 1) a^{11} + 2 b^2 (b^{20} + 45 b^{18} + 307 b^{16} \\ & + 1003 b^{14} + 1887 b^{12} + 2184 b^{10} + 1700 b^8 + 887 b^6 + 292 b^4 + 45 b^2 + 1) a^{10} + 2 b^2 (b^{20} \\ & - 37 b^{18} - 12 b^{16} + 729 b^{14} + 2350 b^{12} + 3184 b^{10} + 2163 b^8 + 613 b^6 - 27 b^4 - 37 b^2 + 1) a^{14} \\ & + b (b^{22} - 47 b^{20} - 77 b^{18} + 36 b^{16} + 585 b^{14} + 992 b^{12} + 192 b^{10} - 733 b^8 - 304 b^6 + 95 b^4 \\ & + 29 b^2 - 1) a^{13} + a^{24} b^{12} + b^{12}) : \end{aligned}$$

#Define r4

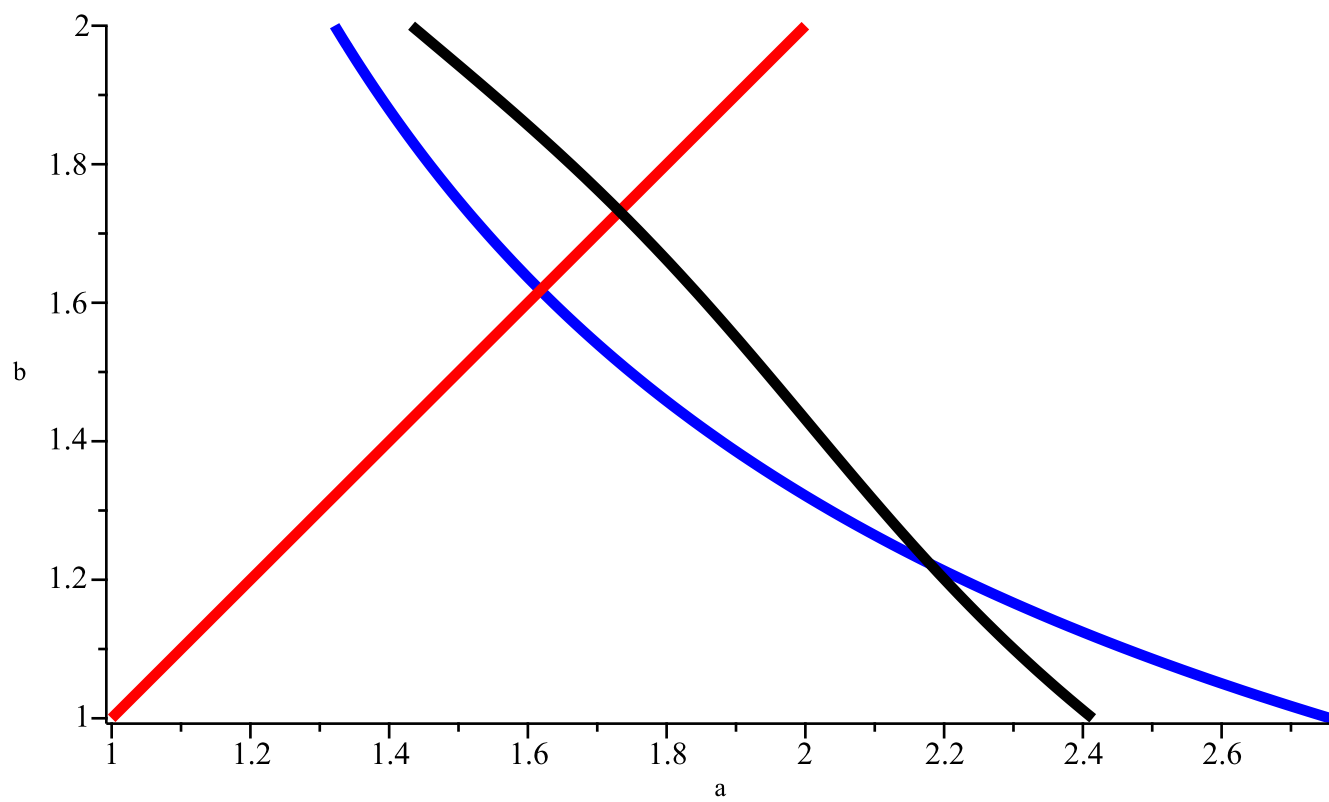
$$\begin{aligned} r4 := & (b^{12} a^{22} + (8 b^{13} - 6 b^{11}) a^{21} + (30 b^{14} + 6 b^{12} + 15 b^{10}) a^{20} + (74 b^{15} + 162 b^{13} - 102 b^{11} \\ & - 18 b^9) a^{19} + (145 b^{16} + 600 b^{14} + 93 b^{12} + 198 b^{10} + 3 b^8) a^{18} + (252 b^{17} + 1380 b^{15} \\ & + 1884 b^{13} - 762 b^{11} - 96 b^9 + 24 b^7) a^{17} + (145 b^{18} + 1092 b^{16} + 1908 b^{14} - 2920 b^{12} \\ & - 3162 b^{10} - 1740 b^8 - 294 b^6) a^{16} + (74 b^{19} + 1380 b^{17} + 4692 b^{15} + 3340 b^{13} - 8010 b^{11} \\ & - 2472 b^9 - 132 b^7 + 24 b^5) a^{15} + (30 b^{20} + 600 b^{18} + 1908 b^{16} - 3516 b^{14} - 24921 b^{12} \\ & - 26352 b^{10} - 11586 b^8 - 1740 b^6 + 3 b^4) a^{14} + (8 b^{21} + 162 b^{19} + 1884 b^{17} + 3340 b^{15} \\ & - 8268 b^{13} - 34416 b^{11} - 14520 b^9 - 2472 b^7 - 96 b^5 - 18 b^3) a^{13} + b^2 (b^{20} + 6 b^{18} + 93 b^{16} \\ & - 2920 b^{14} - 24921 b^{12} - 68232 b^{10} - 67416 b^8 - 26352 b^6 - 3162 b^4 + 198 b^2 + 15) a^{12} \\ & - 6 b (b^{20} + 17 b^{18} + 127 b^{16} + 1335 b^{14} + 5736 b^{12} + 11692 b^{10} + 5736 b^8 + 1335 b^6 + 127 b^4 \\ & + 17 b^2 + 1) a^{11} + (15 b^{20} + 198 b^{18} - 3162 b^{16} - 26352 b^{14} - 67416 b^{12} - 68232 b^{10} \\ & - 24921 b^8 - 2920 b^6 + 93 b^4 + 6 b^2 + 1) a^{10} + (-18 b^{19} - 96 b^{17} - 2472 b^{15} - 14520 b^{13} \\ & - 34416 b^{11} - 8268 b^9 + 3340 b^7 + 1884 b^5 + 162 b^3 + 8 b) a^9 + 3 b^2 (b^{16} - 580 b^{14} \\ & - 3862 b^{12} - 8784 b^{10} - 8307 b^8 - 1172 b^6 + 636 b^4 + 200 b^2 + 10) a^8 + (24 b^{17} - 132 b^{15} \\ & - 2472 b^{13} - 8010 b^{11} + 3340 b^9 + 4692 b^7 + 1380 b^5 + 74 b^3) a^7 + (-294 b^{16} - 1740 b^{14} \\ & - 3162 b^{12} - 2920 b^{10} + 1908 b^8 + 1092 b^6 + 145 b^4) a^6 + (24 b^{15} - 96 b^{13} - 762 b^{11} \\ & + 1884 b^9 + 1380 b^7 + 252 b^5) a^5 + (3 b^{14} + 198 b^{12} + 93 b^{10} + 600 b^8 + 145 b^6) a^4 + (-18 b^{13} - 102 b^{11} + 162 b^9 + 74 b^7) a^3 + (15 b^{12} + 6 b^{10} + 30 b^8) a^2 + (-6 b^{11} + 8 b^9) a \\ & + b^{10}) : \end{aligned}$$

$r := a - b :$

with(plots) :

```
implicitplot([r4=0, r=0, m0=0], a=1 .. 3, b=1 .. (2),
  scaling=constrained,
  gridrefine=2,
  thickness=4,
  color=[blue, red, black, black],
  labels=["a", "b"],
  title="Graphic for r4=0 and m0=0");
```

Graphic for $r_4=0$ and $m_0=0$



#Approximate value for $\gamma_4\left(\frac{3}{2}\right)$

fsolve $\left(\text{eval}\left(r_4, b = \frac{3}{2}\right) = 0, a, 1 \dots 3\right)$

1.747629723

(6)

with(plots) :

implicitplot $\left(\left[r_4 = 0, r = 0, m_0 = 0, \frac{3}{2} - b = 0, 1.747629723 - a = 0\right], a = 1 \dots 3, b = 1 \dots (2),\right.$

$\text{scaling} = \text{constrained},$

$\text{gridrefine} = 2,$

$\text{thickness} = 3,$

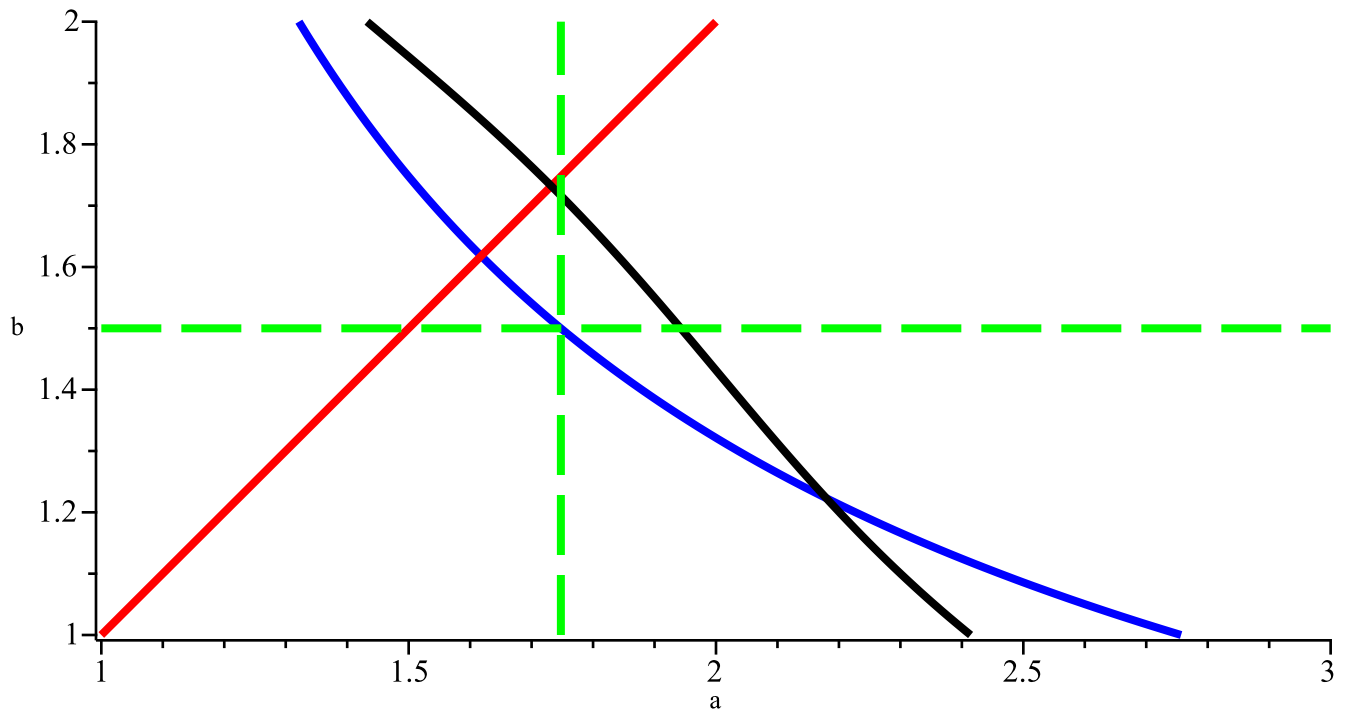
$\text{color} = [\text{blue}, \text{red}, \text{black}, \text{green}, \text{green}],$

$\text{linestyle} = [1, 1, 1, 3, 3],$

$\text{labels} = ["a", "b"],$

$\text{title} = \text{"Graphic for } r_3=0 \text{ and } m_0=0\text{"})$;

Graphic for $r_3=0$ and $m_0=0$



#Locates a range for $\gamma_4\left(\frac{3}{2}\right)$

sturm $\left(\text{sturmseq}\left(\text{eval}\left(r_4, b = \frac{3}{2}\right), a\right), a, 1.5, 1.8\right)$

1

(7)

$$\# \frac{3}{2} < \gamma_4\left(\frac{3}{2}\right) < \frac{9}{5}$$

#Calculate $m_0\left(a, \frac{3}{2}\right)$ and its derivative

eval $\left(m_0, b = \frac{3}{2}\right);$

$$\begin{aligned}
& \text{diff}\left(\text{eval}\left(m0, b = \frac{3}{2}\right), a\right) \\
& \frac{885735}{8192} a - \frac{1476225}{8192} a^2 + \frac{6646474835397}{2097152} a^{14} - \frac{18411713883}{65536} a^{15} + \frac{492215074191}{524288} a^{16} \\
& - \frac{122961235545}{524288} a^{17} + \frac{2468375775}{32768} a^{18} - \frac{15377337189}{131072} a^{19} + \frac{85140306721295}{16777216} a^{12} \\
& + \frac{9345742271613}{2097152} a^{10} - \frac{540160569}{16384} a^{20} - \frac{345259503}{16384} a^{21} - \frac{1476225}{8192} a^{22} \\
& + \frac{111620111427}{65536} a^8 - \frac{540160569}{16384} a^4 + \frac{20429734383}{131072} a^6 - \frac{14862901815}{524288} a^7 \\
& - \frac{63590850603}{524288} a^9 + \frac{118747539}{4096} a^3 + \frac{10101623967}{131072} a^5 + \frac{627868155111}{8388608} a^{13} \\
& - \frac{2685714498807}{8388608} a^{11} + \frac{531441}{4096} - \frac{885735}{8192} a^{23} + \frac{531441}{4096} a^{24} \\
& \frac{885735}{8192} - \frac{1476225}{4096} a + \frac{46525323847779}{1048576} a^{13} - \frac{276175708245}{65536} a^{14} + \frac{492215074191}{32768} a^{15} \quad (8) \\
& - \frac{2090341004265}{524288} a^{16} + \frac{22215381975}{16384} a^{17} - \frac{292169406591}{131072} a^{18} \\
& + \frac{255420920163885}{4194304} a^{11} + \frac{46728711358065}{1048576} a^9 - \frac{2700802845}{4096} a^{19} - \frac{7250449563}{16384} a^{20} \\
& - \frac{16238475}{4096} a^{21} + \frac{111620111427}{8192} a^7 - \frac{540160569}{4096} a^3 + \frac{61289203149}{65536} a^5 \\
& - \frac{104040312705}{524288} a^6 - \frac{572317655427}{524288} a^8 + \frac{356242617}{4096} a^2 + \frac{50508119835}{131072} a^4 \\
& + \frac{8162286016443}{8388608} a^{12} - \frac{29542859486877}{8388608} a^{10} - \frac{20371905}{8192} a^{22} + \frac{1594323}{512} a^{23}
\end{aligned}$$

#Verify that the derivative of $m0\left(a, \frac{3}{2}\right)$ does **not** vanish in the interval $[1.5, 1.8]$

$$\text{ SturmSeq}\left(\text{diff}\left(\text{eval}\left(m0, b = \frac{3}{2}\right), a\right), a\right), a, \frac{3}{2}, 1.8 \Big)$$

0

(9)

#Verify that $m0\left(\frac{3}{2}, \frac{3}{2}\right)$ and $m0\left(\frac{9}{5}, \frac{3}{2}\right)$ are positive

$$\text{eval}\left(\text{eval}\left(m0, b = \frac{3}{2}\right), a = \frac{3}{2}\right); \text{eval}\left(\text{eval}\left(m0, b = \frac{3}{2}\right), a = \frac{9}{5}\right)$$

60713027138609389089
34359738368

(10)

#This concludes the existence of a point $(\gamma_4(b), b)$ with $m_0 = 0$

```
#Compute the resultant R1
```

$$bfator := [op(factor(R1))] : \\ nops(bfator);$$

$$bfator[1] = -28991029248 \quad (12)$$

$$bfator[4] \quad (b^2 + 1)^{162} \quad (14)$$

$$\text{degree}(\text{bfator}[3], b)$$

204 **(15)**

$$seq\left(sign\left(evalf\left(coeff\left(numer\left(eval(bfactor[3], b = \frac{1 + 2d}{1 + d})\right), d, i\right)\right), i = 204 .. 0, -1\right);$$

#The factor $bfator[3]$ has at most a unique root between 1 and 2

```
fsolve(bfator[3]=0, b, 1..2);
```

$$1.223054666 \quad (17)$$

#Compute the resultant R2

$R2 := \text{resultant}(r4, m0, b) :$

$afator := [op(factor(R2))]:$
 $nops(afator);$

4 (18)

$afator[1]$
 -28991029248 (19)

$afator[2]$
 a^{264} (20)

$afator[4]$
 $(a^2 + 1)^{162}$ (21)

#The third factor is highly significant.

$degree(afator[3], a)$
 204 (22)

#Count the number of zeros in $afator[3]$ in the interval $[1, 3]$

$seq\left(sign\left(evalf\left(coeff\left(numer\left(eval\left(afator[3], a = \frac{1+3c}{1+c}\right)\right), c, i\right)\right), i = 204 .. 0, -1\right);$

-1, (23)
 -1,
 -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
 1,
 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1,
 1, -1,
 -1,
 -1,
 -1, -1, -1, -1

#The factor $afator[3]$ has at most four roots between 1 and 3

#Finding possible candidates for zeros of the factor $afator[3]$

$fsolve(afator[3] = 0, a, 1 .. 3);$
 1.216407333, 1.453232550, 1.530050281, 2.179476594 (24)

#Thus, the candidate points (a,b) for the intersection of $r_3=0$ and $m_0=0$ are of the form $a \in A$ and $b \in B$ where

$A := [1.216407333, 1.453232550, 1.530050281, 2.179476594] :$

$B := [1.223054666] :$

$Points := [seq(seq([a, b], b \in B), a \in A)] :$

$Points_filtered := select(p \rightarrow p[2] < p[1], Points);$

$Points_filtered := [[1.453232550, 1.223054666], [1.530050281, 1.223054666], [2.179476594, 1.223054666]]$ (25)

with(plots) :

with(plottools) :

```
p1 := implicitplot(
  [r4 = 0, r = 0],
  a = 1 .. 3, b = 1 .. 2,
  scaling = constrained,
  gridrefine = 2,
  thickness = 2,
  color = [blue, red],
  labels = ["a", "b"],
  title = "Graphic for r1=0 in A1"
) :
```

rectangles

```
R1 := rectangle([1, 1], [1.6, 1.4],
  color = silver, transparency = 0.5,
  linestyle = 3, # pontilhado
  thickness = 2) :
```

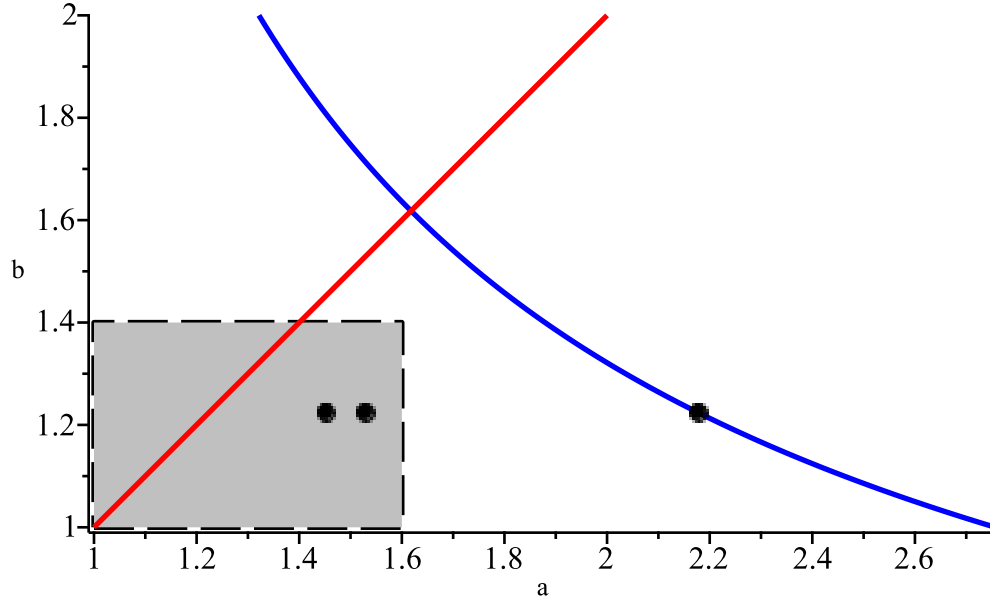
points

```
P := pointplot(
  Points_filtered,
  symbol = solidcircle,
  symbolsize = 12,
  color = black
) :
```

sobrepor tudo

```
display([p1, P, R1]);
```

Graphic for r1=0 in A1



#By covering the previous points with rectangles, we will show that the curve $r4=0$ does not intersect them.

restart;

#Define $r4$

$$\begin{aligned}
 r4 := & (b^{12} a^{22} + (8 b^{13} - 6 b^{11}) a^{21} + (30 b^{14} + 6 b^{12} + 15 b^{10}) a^{20} + (74 b^{15} + 162 b^{13} - 102 b^{11} \\
 & - 18 b^9) a^{19} + (145 b^{16} + 600 b^{14} + 93 b^{12} + 198 b^{10} + 3 b^8) a^{18} + (252 b^{17} + 1380 b^{15} \\
 & + 1884 b^{13} - 762 b^{11} - 96 b^9 + 24 b^7) a^{17} + (145 b^{18} + 1092 b^{16} + 1908 b^{14} - 2920 b^{12} \\
 & - 3162 b^{10} - 1740 b^8 - 294 b^6) a^{16} + (74 b^{19} + 1380 b^{17} + 4692 b^{15} + 3340 b^{13} - 8010 b^{11} \\
 & - 2472 b^9 - 132 b^7 + 24 b^5) a^{15} + (30 b^{20} + 600 b^{18} + 1908 b^{16} - 3516 b^{14} - 24921 b^{12} \\
 & - 26352 b^{10} - 11586 b^8 - 1740 b^6 + 3 b^4) a^{14} + (8 b^{21} + 162 b^{19} + 1884 b^{17} + 3340 b^{15} \\
 & - 8268 b^{13} - 34416 b^{11} - 14520 b^9 - 2472 b^7 - 96 b^5 - 18 b^3) a^{13} + b^2 (b^{20} + 6 b^{18} + 93 b^{16} \\
 & - 2920 b^{14} - 24921 b^{12} - 68232 b^{10} - 67416 b^8 - 26352 b^6 - 3162 b^4 + 198 b^2 + 15) a^{12} \\
 & - 6 b (b^{20} + 17 b^{18} + 127 b^{16} + 1335 b^{14} + 5736 b^{12} + 11692 b^{10} + 5736 b^8 + 1335 b^6 + 127 b^4 \\
 & + 17 b^2 + 1) a^{11} + (15 b^{20} + 198 b^{18} - 3162 b^{16} - 26352 b^{14} - 67416 b^{12} - 68232 b^{10} \\
 & - 24921 b^8 - 2920 b^6 + 93 b^4 + 6 b^2 + 1) a^{10} + (-18 b^{19} - 96 b^{17} - 2472 b^{15} - 14520 b^{13} \\
 & - 34416 b^{11} - 8268 b^9 + 3340 b^7 + 1884 b^5 + 162 b^3 + 8 b) a^9 + 3 b^2 (b^{16} - 580 b^{14}
 \end{aligned}$$

[illegible]